

Course ID: SW 302

User Interface (UI) Development

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Lecture#2

Introduction to HTML

Lecture Outline

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- What is HTML?
- Is HTML a Programming Language?
- HTML Tags
- What is HTML5?
- HTML5 Built-in Validation

What is HTML?

What is HTML?

HTML, otherwise known as **HyperText Markup Language**, is the language used to create Web pages.

Using **HTML**, you can create a Web page with text, graphics, sound, and video.

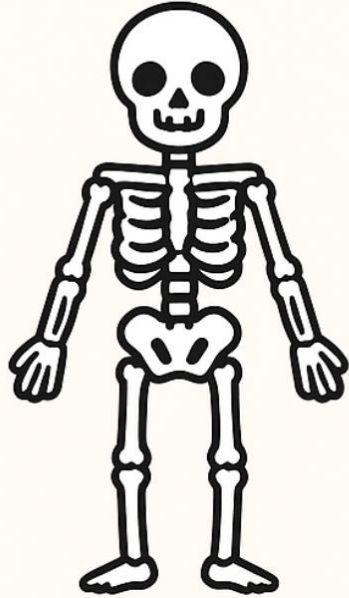


HyperText Markup Language (HTML)

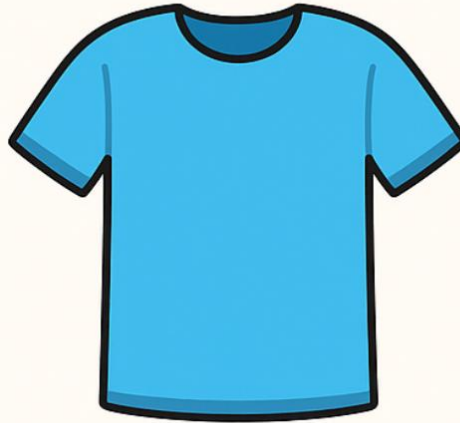
[.āch-(.)tē-(.)em-'el]

The basic scripting language used by web browsers to render pages on the world wide web.

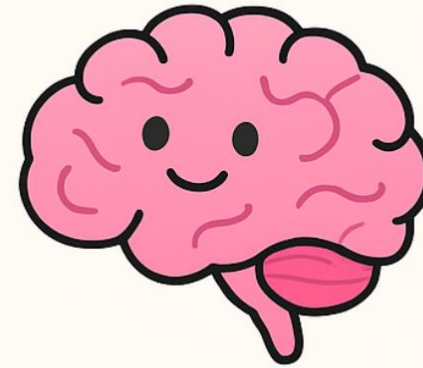
HTML vs CSS vs JavaScript



HTML
Skeleton
(structure)



CSS
Clothes
(style)



JavaScript
Muscles/Brain
(behavior & interactivity)

Is HTML a Programming Language?

HTML is not a programming language!?

Not a programming language. (e.g., C++, Python, or Java)

Not a scripting language (e.g., Batch scripts, shell scripts)

It is a markup language

It does **not contain logic**, such as conditions, loops, or variables—that's what programming/scripting languages do.

HTML Purpose: **defines the structure and content** of a webpage (headings, paragraphs, images, links, etc.)

HTML Tags

HTML is not a programming language.

- **Tags** are the **building blocks of HTML**
- They define **elements** on a webpage (headings, paragraphs, images, links, etc.).
- Written inside **angle brackets < >**.
- Usually come in **pairs**:
 - Opening tag <p> and closing tag </p>
- Some tags are **self-closing**: ,

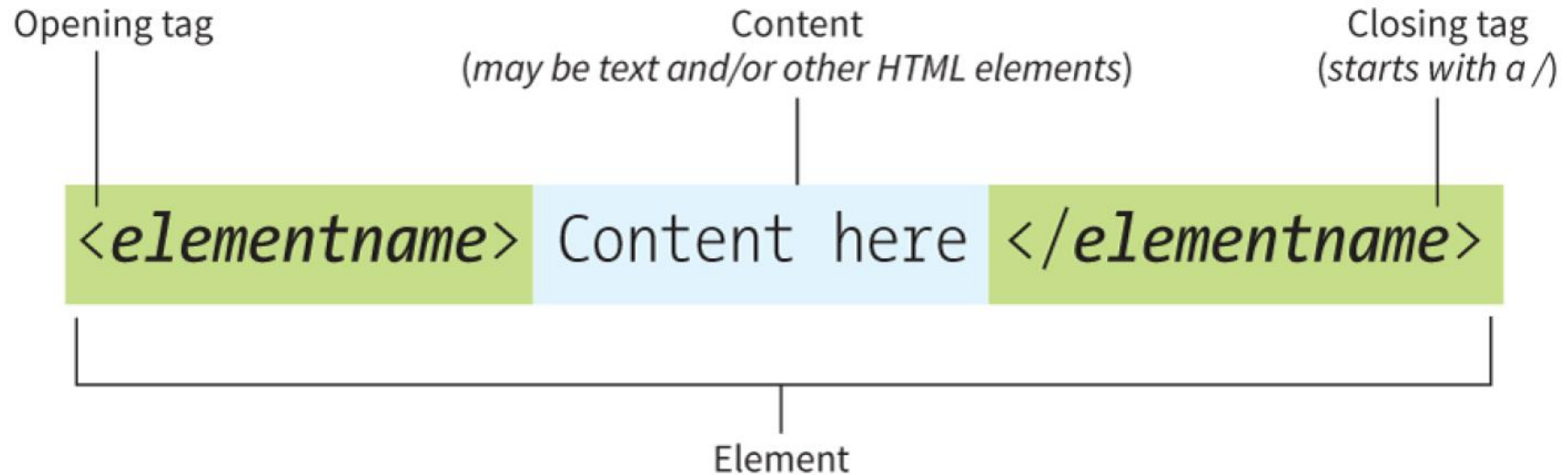
Anatomy of an HTML Element

tag

The element name in angle brackets

element

The content and its markup (start and end tags)



Example:

`<h1>Black Goose Bistro</h1>`

Types of HTML Tags

- **Basic Structure Tags:** <html>, <head>, <body>.
- **Text Tags:** <h1> to <h6>, <p>, , <i>.
- **Links & Media:** <a>, , <video>, <audio>.
- **Lists:** , ,
- **Tables:** <table>, <tr>, <td>

Nested Tags

Whenever you have HTML tags within other HTML tags, you must close the nearest tag first

Nested tags = placing one HTML element **inside another**

They create a hierarchy or structure of the web page. **Must follow a proper opening and closing order (like a stack).**

**Which one
is correct?**

Example#1

```
<p><i>This text is italic and  
<b>bold</b></i></p>
```

Example#2

```
<p><i>This text is italic and  
<b>bold</i></b></p>
```

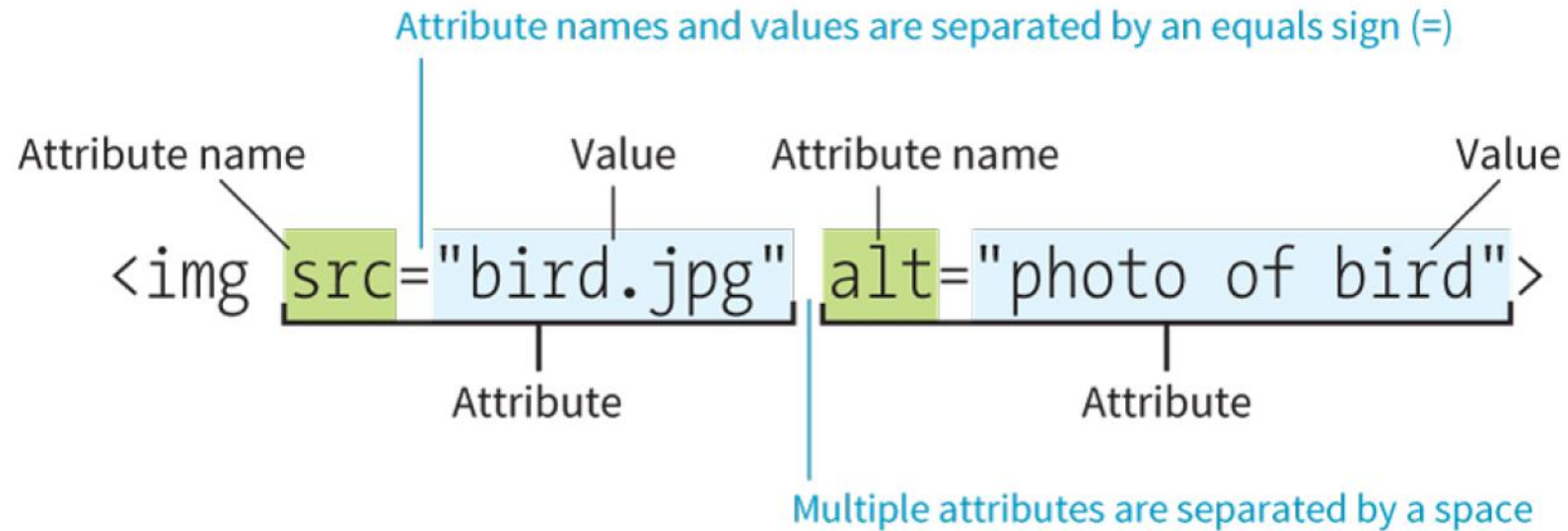
Attributes

Attributes are instructions that clarify or modify an element. They appear in the opening tag after the element name:

```
<element attribute="value">Content</element>
```

Attributes

There can be more than one attribute in a tag:



They are separated by spaces and can go in any order.

Structure of a Web HTML Page

- All Web pages share a common structure
- All Web pages should contain a pair of `<HTML>`, `<HEAD>`, `<TITLE>`, and `<BODY>` tags

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```
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    <!-- Page content goes here.-->
  </body>
</html>
```

Structure of a Web HTML Page

The html element is the **root element** that contains all the elements in the document:

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```
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Structure of a Web HTML Page

The **head** element contains elements that pertain to the document and are not rendered as content, such as title, metadata, style sheets, and scripts

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```
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Structure of a Web HTML Page

Meta elements provide document **metadata**. In this case, it says that the document uses an expanded **character set**(UTF-8). It could also provide keywords, author information, publishing status, and more

```
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Structure of a Web HTML Page

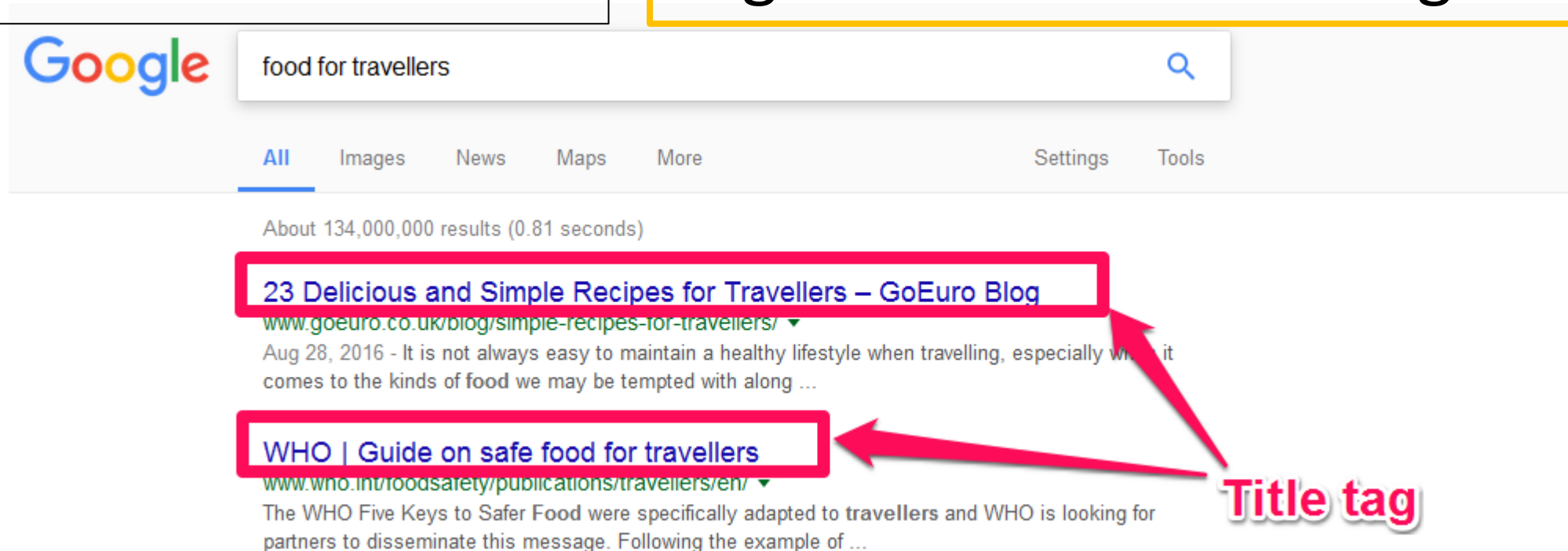
The **title** element is required. Titles display in the browser tab and bookmark lists, and are used by search engines:

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Title here</title>
  </head>
  <body>
    [Page content goes here.]
  </body>
</html>
```

Structure of a Web HTML Page

The title of a Web page determines its ranking in certain search engines

The content of the <title> tag **appears in the browser tab/history** and is used by search engines and bookmarking tools.

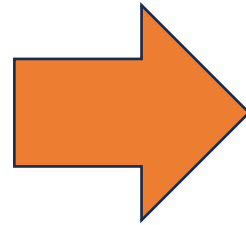


HTML Hello World

```
<!DOCTYPE html>
<html>
<!-- head tag -->

<head>
  <title>Welcome to SW 302 </title>
</head>
<!-- Body tag -->

<body>
  <h2> SW 302 </h2>
  <p>
    A Computer Science Portal for SW 302
  </p>
  <p>
    Please change the code as you like.
  </p>
</body>
</html>
```



SW 302

A Computer Science Portal for SW 302

Please change the code as you like.

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Text Formatting Tags

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 Bold Face

<I> *Italics* </I>

<U> Underline </U>

<P> New Paragraph </P>

 Next Line

Changing the Font

The expression ** ... ** can be used to change the font of the enclosed text

To change the size of text use the expression ** ** where n is a number between 1 and 7



Headings

- Heading tags define headings in HTML.
- They range from `<h1>` (most important) to `<h6>` (least important).
- Headings give structure and hierarchy to a webpage.

Think of headings like an **outline in a textbook**:

`<h1>` = Book Title

`<h2>` = Chapter Title

`<h3>` = Section Title inside the chapter

Headings

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

```
<h1>Heading 1</h1>
```

```
<h2>Heading 2</h2>
```

```
<h3>Heading 3</h3>
```

```
<h4>Heading 4</h4>
```

```
<h5>Heading 5</h5>
```

```
<h6>Heading 6</h6>
```

Aligning Text

- The ALIGN attribute can be inserted in the <Hn> tags to **right-justify**, center, or **left-justify** the text
- For example, <H1 **ALIGN=CENTER**> The New York Times </H1> would create a centered heading of the largest size

center

left 2

right 3

justify

Comment Statements

- The comments do not appear on the Web page itself but are a useful reference to the author of the page and other programmers
- To create a comment statement use the **<!-- -->** tags

```
1 <html>
2
3 <body>
4
5 <h1> Hi, How are you?</h1>
6 <br>
7 <!-- Sample HTML comment -->
8 <!-- This two lines are not displayed on the browser -->
9
10 <h2>Welcome to www.TestingDocs.com</h2>
11
12 </body>
13
14 </html>
15
16
17
```



HTML Comments

Page Formatting

- To define the background color, use the **BGCOLOR** attribute in the **<BODY>** tag
- To define the text color, use the **TEXT** attribute in the **<BODY>** tag

```
<HTML>
```

```
<HEAD>
```

```
<TITLE> Example </TITLE>
```

```
</HEAD>
```

```
<BODY BGCOLOR="black" TEXT="white">
```

This is where you would include the text and images on your Web page.

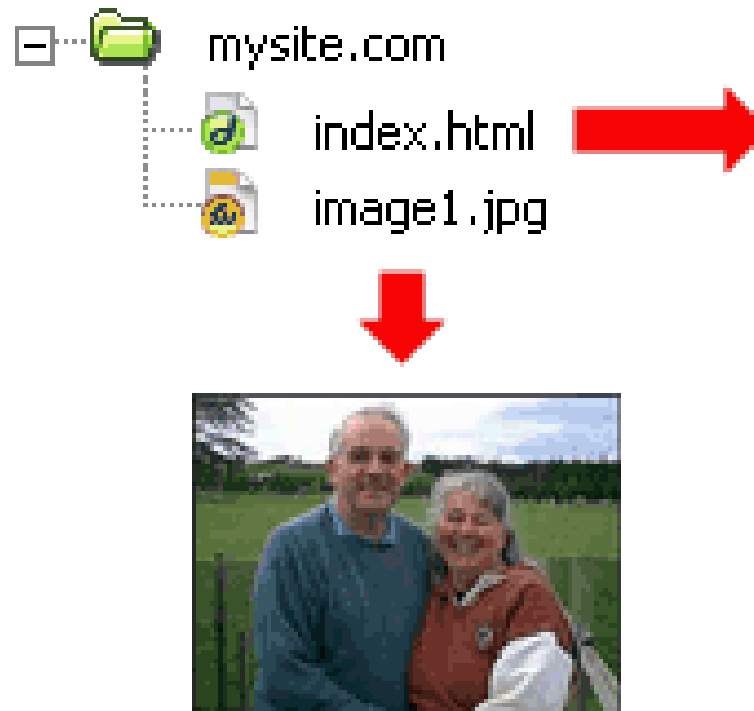
```
</BODY>
```

```
</HTML>
```

Inserting Images

- Type **, where image.ext indicates the location of the image file
- The **WIDTH=n** and **HEIGHT=n** attributes can be used to adjust the size of an image
- The attribute **BORDER=n** can be used to add a border n pixels thick around the image

HTML and Image File



```
<html>
<head>
<title>Picture Page</title>
</head>
<body>
Here's a picture:<br>

</body>
</html>
```

Alternate Text

- **Accessibility:** Screen readers read alt text aloud for visually impaired users. Ensures everyone can understand the image's meaning.
- **SEO (Search Engine Optimization):**
 - Search engines use alt text to understand what the image is about.
 - Helps images appear in Google Images search.
- **Fallback:** If an image doesn't load, the alt text is displayed instead.

Example:

```
<IMG SRC="satellite.jpg" ALT = "Picture of satellite">
```

Links

- The `<a>` tag is used to create hyperlinks. “a” stands for anchor.
- It connects one page (or part of a page) to another resource: a web page, a file, an email, or even a location within the same page.

1. Link to a Website

```
<a href="https://www.google.com">Visit Google</a>
```

2. Page in Your Site

```
<a href="about.html">About Us</a>
```

3. Open in New Tab

```
<a href="https://www.github.com"  
target="_blank">Visit GitHub</a>
```

4. Link to Email

```
<a href="mailto:info@example.com">Email Us</a>
```

Ordered Lists

► Ordered lists are a list of numbered items.

► To create an ordered list, type:

```
<OL>
```

```
<LI> This is step one.
```

```
<LI> This is step two.
```

```
<LI> This is step three.
```

```
</OL>
```

1. This is step one.
2. This is step two.
3. This is step three.

The **TYPE=x** attribute allows you to change the kind of symbol that appears in the list.

A is for capital letters

a is for lowercase letters

I is for capital Roman numerals

i is for lowercase roman numerals

Unordered Lists

► An unordered list is a list of bulleted items

► To create an unordered list, type:

```
<UL>
```

```
  <LI> First item in list
```

```
  <LI> Second item in
```

list

```
  <LI> Third item in list
```

```
</UL>
```

- First item in list
- Second item in list
- Third item in list

The **TYPE=shape** attribute allows you to change the type of bullet that appears

circle corresponds to an empty round bullet

square corresponds to a square bullet

disc corresponds to a solid round bullet; this is the default value

Div Tag

- Use the div element to create a logical grouping of content or elements on the page . Indicates division of content (generally block level) - Useful for creating “hooks” for scripts and style rules.
- It indicates that they belong together in some sort of conceptual unit or should be treated as a unit by CSS or JavaScript.

```
<div class="listing">
```

```
  
```

```
  <p><cite>The Complete Manual of Typography</cite>, James  
  Felici</p>
```

```
  <p>A combination of type history and examples of good and  
  bad type design.</p>
```

```
</div>
```

Span Tag

► Use the span element for text and other inline elements for which no existing inline element currently exists.

- Useful for creating “hooks” for scripts and style rules.

► In this example, a span is used to add semantic meaning to telephone numbers:

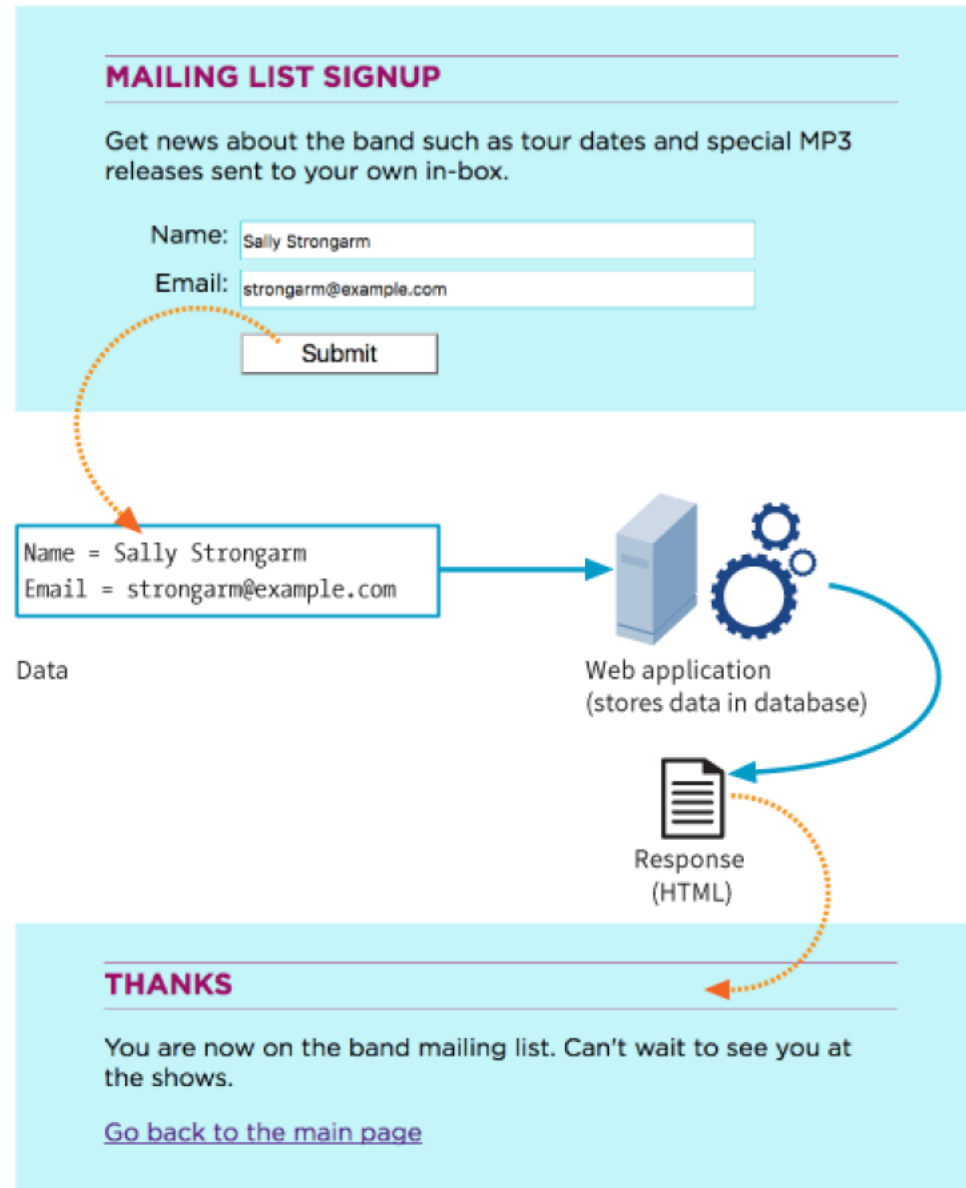
```
<ul>
  <li>John: <span class="tel">999.8282</span></li>
  <li>Paul: <span class="tel">888.4889</span></li>
  <li>George: <span class="tel">888.1628</span></li>
  <li>Ringo: <span class="tel">999.3220</span></li>
</ul>
```

Web Forms

1. Browser renders the form inputs as indicated in the markup.

2. User enters information in the form and hits Submit.

3. The browser encodes the information entered and sends it to the server.



4. The application processes the information.

5. The application returns a response (for example, a thank you message or reloading the page).

Server Side Processing

PHP (.php)
Microsoft ASP (.asp)
Microsoft ASP.net (.aspx)
Ruby on Rails
JavaServer Pages (.jsp)
Python

Forms Tag

► The **<form>** tag is one of the most important parts of HTML because it's how users **send data** to a server (e.g., login, signup, search, payment, etc.).

action → URL of the server or script that processes the data.

method → How data is sent:

- **"get"** → Data is appended to the URL (visible in the browser).
- **"post"** → Data is sent hidden in the HTTP body (more secure).

```
<form action="server_page.php"
method="post">
  <!-- form controls go here -->
</form>
```

Forms Tag : More on method attribute

```
<form action="mailinglist.php" method="POST">
```

The **method** attribute specifies how the encoded information should be sent to the server (GET or POST):

GET: The encoded data is tacked onto the URL sent to the server:

```
get  
http://www.bandname.com/mailinglist.php?name=Sally%20Strong  
arm&email=strongarm%40example.com
```

POST: Data is sent in a separate transaction and can be encrypted with HTTPS.

NOTE: POST is the most common method.

Common Form Elements

- **<input type="text">** → single-line text.
- **<input type="password">** → password field.
- **<input type="email">** → email input (with validation).
- **<input type="radio">** → radio button.
- **<input type="checkbox">** → checkbox.
- **<input type="submit">** → submit button.
- **<textarea>** → multi-line text
- **<select>** and **<option>** → dropdown menus.
- **<button>** → clickable button.

INPUT TYPES IN HTML

`<input type = "text">`

xyz

`<input type = "password">`

`<input type = "radio">`

No ☐ Yes

`<input type = "checkbox">`

☐ ☐ ☒

`<input type = "button">`

Button

`<input type = "email">`

Enter email

`<input type = "file">`

Choose file

`<input type = "hidden">`

`<input type = "image">`

 Submit Image

`<input type = "number">`

1234

`<input type = "range">`

`<input type = "search">`

Search

`<input type = "tel">`

123-456-789

`<input type = "time">`

12:44 AM

`<input type = "date">`

12-11-22 

Form - Example

First Name: ← Text Box

Last Name:

Type of Shirt: ← Drop-down Menu

Size: ☐ Large ☒ Medium ☐ Small ← Radio Buttons

Color: ☐ Red ☒ Navy ☐ Black ← Checkboxes

Comments?

← Text Area

Reset Button

Submit Button

Creating Submit / Reset Button

- To create a submit button, type **<INPUT TYPE="submit">**.
- If you would like the button to say something other than submit, use the **VALUE** attribute
- **For example:** **<INPUT TYPE="submit" VALUE="Buy Now!">** would create a button that says "Buy Now!"
- To create a **reset** button, type **<INPUT TYPE="reset">**

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HTML **<input type="submit">**

User Name:

Password:

Submit

Creating Text Input Boxes

- To create a text box, type `<INPUT TYPE="text" NAME="name" VALUE="value" SIZE=n MAXLENGTH=n>`
- The NAME, VALUE, SIZE, and MAXLENGTH attributes are optional

The **NAME** attribute is used to identify the text box to the processing script

The **VALUE** attribute is used to specify the text that will initially appear in the text box

The **SIZE** attribute is used to define the size of the box in characters

The **MAXLENGTH** attribute is used to define the maximum number of characters that can be typed in the box

Creating Larger Text Areas

A **multi-line text input** control (unlike `<input type="text">` which is single-line).

To create larger text areas, type `<TEXTAREA NAME="name" ROWS=n1 COLS=n2 WRAP>` Default Text `</TEXTAREA>`, where n1 is the height of the text box in rows and n2 is the width of the text box in characters

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Send your comments to the author.

Submit

Drop-down Menu

The **<select>** tag in **HTML** is used to create a drop-down list (also called a selection list)

<select>

<option value="C++">C++</option>

<option

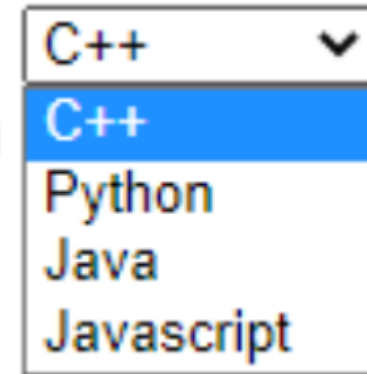
value="Python">Python</option>

<option value="Java">Java</option>

<option

value="Javascript">Javascript</option>

</select>



name → identifies the field when the form is submitted.

id → links the **<select>** to a **<label>**.

size → shows multiple options at once instead of a drop-down (e.g., **size="4"**).

multiple → allows selecting more than one option.

disabled → makes the drop-down unselectable.

required → makes selection mandatory before submitting the form.

Tables

- Tables are made up of cells arranged into rows.

Menu item	Calories	Fat (g)
Chicken noodle soup	120	2
Caesar salad	400	26

table

row	Menu item header cell	Calories header cell	Fat header cell
row	Chicken noodle soup data cell	120 data cell	2 data cell
row	Caesar salad data cell	400 data cell	26 data cell

Tables

- Tables can be used to display rows and columns of data, create multi-column text, captions for images, and sidebars.
- The **<TABLE>** tag is used to create a table; the **<TR>** tag defines the beginning of a row, while the **<TD>** tag defines the beginning of a cell.

Adding a Border

The **BORDER=n** attribute allows you to add a border n pixels thick around the table.

To make a solid border color, use the **BORDERCOLOR="color"** attribute.

To make a shaded colored border, use **BODERCOLORDARK="color"** and **BORDERCOLORLIGHT="color"**

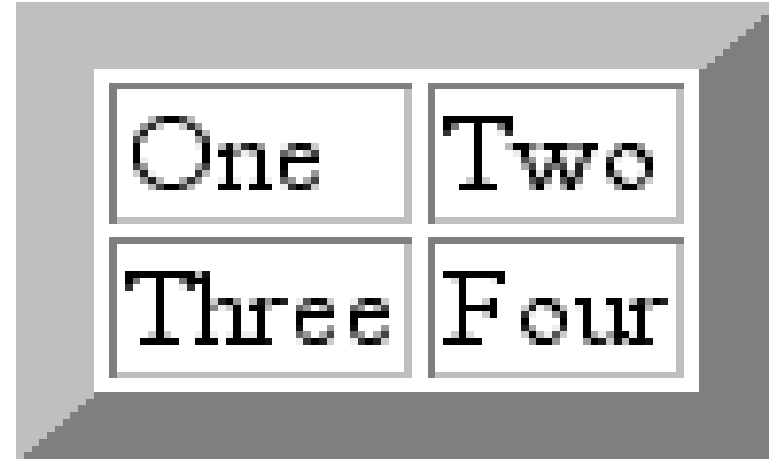
Tables

➤ How it looks using markup (**table**, **tr**, **th**, and **td**):

Menu item	Calories	Fat
Chicken noodle soup	120	2
Caesar salad	400	26

Simple Table - Example

```
<TABLE BORDER=10>
  <TR>
    <TD>One</TD>
    <TD>Two</TD>
  </TR>
  <TR>
    <TD>Three</TD>
    <TD>Four</TD>
  </TR>
</TABLE>
```



Note:

- When a Web browser displays a table, it often adds extra space. To eliminate this space, use the **WIDTH =n** attribute in the **<TABLE>** and **<TD>** tags.
- Keep in mind - a cell cannot be smaller than its contents, and if you make a table wider than the browser window, users will not be able to see parts of it.

Table Headers: <th> </th>

th stands for table header.

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- Headers provide information about the cells in the row or column they precede.
- They are key tools for making table content accessible. Screen readers may read headers aloud before each data cell, providing context that is missing when you can't see the table.
- Headers are often rendered in a bold font by default.

Table Headers: <th> </th>

This table would have **ONLY** two rows and three columns (because there are three cells defined in each row, one of them is a header <td>:

```
<table>
  <tr>
    <th>Burgers</th>
    <td>Organic Grass-fed Beef</td>
    <td>Black Bean Veggie</td>
  </tr>
  <tr>
    <th>Fries</th>
    <td>Hand-cut Idaho potato</td>
    <td>Seasoned sweet potato</td>
  </tr>
</table>
```

What is HTML5?

HTML Evolution

HTML5 was officially finalized in 2014 and continues to evolve. Developed by **W3C (World Wide Web Consortium)**.



Overview - Key Features of HTML5

Semantic Elements – new tags that give meaning to content:

<header>, <footer>, <article>, <section>, <nav>, <aside>

Improves accessibility & SEO.

Multimedia Support – no plugins needed:

<audio> and <video> to play media directly in the browser.

Supports controls, autoplay, captions, etc.

Graphics & Animation::

<canvas> → for 2D graphics, games, data visualization.

<svg> → scalable vector graphics.

Overview - Key Features of HTML5

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Form Enhancements:

New input types like **email, url, date, range, color.**

Attributes like required, placeholder, and autofocus. Improves accessibility & SEO.

APIs for Rich Web Apps:

Geolocation API (get user's location)

Local Storage & Session Storage (store data on the client side).

WebSockets (real-time communication).

Cross-Platform & Mobile Friendly / Designed for responsive design

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Thank You...